



Capstone Project/ Mini Project Progress Journal

**FACULTY OF COMPUTING
SABARAGAMUWA UNIVERSITY OF SRI LANKA**

The second version (September 2024) of the Capstone (IS4110)/Mini Project (SE5104) Project Progress Journal was revised and compiled by:
Faculty of Computing

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Version 2.0 September 2024

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Introduction to Capstone/Mini Projects

Overview

Capstone Project (IS4110)/ Mini Project (SE 5104) is a compulsory course module to complete the Bachelor of Science Honours Degree Programme in Information Systems and Bachelor of Science Honours Degree Programme in Software Engineering offered by the Department of Computing and Information Systems (DCIS) and Department of Software Engineering (DSE), Faculty of Computing, Sabaragamuwa University of Sri Lanka. This is a group project which every Student should have to complete within their relevant Semesters. Information Systems students should complete their Capstone Project in Semester IV and Software Engineering student should complete their Mini Project in Semester V. This course will enable students to carry out project work to implement a solution for real-world IT problem, under the supervision of an internal supervisor and a mentor. The Internal Supervisor provided by the respective department is responsible for helping students if they have any internal matters and they should monitor the interaction between the students. Every group should find their own mentor who is an expert in the industry and those mentors are responsible for supporting and guiding students throughout the projects and evaluate the progress. The mentors should be aware of the students about the emerging technologies and current trends in the Software Industry and guide the students to apply those technologies to their projects. Also, Mentors should monitor the progress of the student continuously and all the students are required to submit a Project Progress Journal once in three weeks to the mentor. Mentors and Internal Supervisors can schedule meetings with the students to assess the progress as well as to discuss the project work. A student is responsible for updating the progress of the project addressing the feedback given by the internal supervisors and mentors throughout the semester and completing the final software product before the end of the semester by covering all the requirements and the scope defined initially. The Capstone Project/Mini Project is evaluated by the internal supervisors and the same panel of evaluators in the Proposal Presentation and the Final Evaluation.

The duration for the Capstone Project/Mini Project is 15 Weeks (One Semester) and it weighted 2 credits. Upon successful completion of this course, the student will be able to identify the requirements for real-world IT problems, study & enhance software/ hardware skills, demonstrate & build the project successfully by hardware requirements, coding, and emulating & testing Report & present the findings of the study conducted in the preferred domain.

Vision

Our vision is to guide the Software Engineering (SE)/Information Systems (IS) students to achieve their individual goals and to lead the next generation of IT professionals in advancing the Information Technology and Business Process Management (IT/BPM) industry.

Mission

Our mission is to produce Software Engineering/Information Systems graduates to design and develop quality software solutions, work effectively within challenging environments, and be good professionals. Since the field of Software Engineering/Information Systems has rapidly grown and diversified, we have to create Software Engineering/Information Systems graduates who have high-quality standards, broad-based education and experiential learning in Software Engineering, create knowledge through pioneering scholarship and impactful research, enrich our students' development and leadership skills, and nurture the inherent innovation of our students.

Objectives

Objectives of the Capstone Project/Mini Project work:

- Offer students a glimpse into real world problems and challenges that need IT based solutions
- Enable students to create very precise specifications of the IT solution to be designed.
- Introduce students to the vast array of literature available of the various research challenges in the field of IT
- Create awareness among the students of the characteristics of several domain areas where IT can be effectively used.
- Enable students to use all concepts of IT in creating a solution for a problem
- Improve the team building, communication and management skills of the students.
- Helps in exploring career opportunities in their areas of interest.
- Gives an insight into the working of the real organizations/companies.
- Develop students' ability to apply knowledge and techniques learnt in theoretical classes to develop software products for real world problems.
- Gives an insight of a real working environment of an organization to the students
- Assist students in exploring career opportunities in their areas of interest.
- Develop students' reporting, presenting and demonstrating ability.

Intended Learning Outcomes

Upon successful completion of this course, the student will be able to:

- ILO1 Discover potential research/Development areas in the field of IT
- ILO2 Conduct a survey of several available literature in the preferred field of study.
- ILO3 Compare and contrast the several existing solutions for research/development challenge
- ILO4 Demonstrate an ability to work in teams and manage the conduct of the research study or Development project
- ILO5 Formulate and propose a plan for creating a solution for the research/development plan identified
- ILO6 To report and present the findings of the study conducted in the preferred domain

Introduction to Project Progress Journal

Project Progress Journal is

- A way to document all Capstone/Mini Project activities throughout the semester.
- An important mechanism for the internal supervisors and mentor to evaluate and assess a student's attitude and ability and also to monitor the status of the student's project throughout the semester.

Students are

- Required to write clearly and honestly all activities performed and then to summarize their work every week.
- Highly encourage to maintain a separate file/folder to compile all their findings/ printouts/ datasheets as a complement to this progress journal.

Reminder to students

- This journal must be presented to your Mentor and the Internal Supervisor once in three weeks.
- All activities conducted must be recorded at the activities section in the Project Progress Journal. Signatures of relevant persons can also be recorded as proof of your claim at the activities section (optional but highly recommended).
- This journal must be submitted to your internal supervisor and mentor along with your Final Report to be graded by your internal supervisor and the mentor at the required date (Refer to Capstone/Mini Project Guidelines).

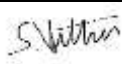
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Student Information

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Mentors' Details	Name	
	Profession	
	Contact Details	
Project Title	Smart UniMate – University Chatbot and Student Engagement Platform	
Project Description	Smart UniMate is a digital assistant service that works 24/7 and is specifically designed to assist new university students in adjusting to campus life as smoothly as possible. The core of the digital assistant is a chatbot based on Retrieval-Augmented Generation (RAG), which provides swift and verified information from the university's official documents. The digital assistant also includes a student community section that is backed by Azure Active Directory (AAD) security, allowing users to assist each other. The end objective is to reduce the complexities faced during the orientation process and provide all the information a student needs in a single place.	

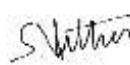
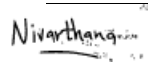
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Week 1

Duration	Start Date	16.03.2026	End Date	22.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
16.02.2026	Conducted initial team meeting to discuss our project idea	Establish a project workflow for project execution and define individual roles within the team.	None (Already discussed during proposal)	None
17.02.2026-18.02.2026	Held consultation session with internal supervisor to review the proposed system scope.	Validate the feasibility and scope of the project.	Some proposed features exceeded the expected project scope.	Improved key system features in response to input.
19.02.2026	None	None	None	None
20.02.2026	Performed detailed review of the proposal and refined the requirements elicitation strategy.	Ensure alignment between the proposed solution and expected system outcomes.	Certain requirements had low clarity.	Conducted internal discussion to clarify the requirements.
21.02.2026	Initiated requirements elicitation through discussions with undergraduates and categorized collected required data.	Identify user expectations and common challenges faced by university freshers.	Large amount of raw feedback data.	Applied requirement classification techniques.
22.02.2026	Installed and set up the wanted programs.	To work on the project	Found difficulties to download some files.	Watched videos to get clarity.
Summary for the week	This week, after the project proposal was successfully submitted, the project team concentrated on creating an organized development workflow. Meetings with the internal supervisor yielded insightful input on the viability and extent of the suggested system. To better understand the demands of potential users, especially first-year university students, preparatory requirement elicitation activities were also carried out.			
Task for the next week	Conduct detailed requirement analysis and studying Figma prototyping features.			
Remarks				
Signature of the Student			Date	16.03.2026
Signature of the Mentor			Date	16.03.2026

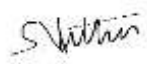
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Week 2

Duration	Start Date	23.02.2026	End Date	01.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
23.02.2026	Conducted requirement analysis based on the collected data.	Identify functional and non-functional system requirements.	Some requirements overlapped across system modules.	Applied requirement prioritization.
24.02.2026	Started learning Figma (Watched Figma tutorials to learn basic objectives)	To learn the basic of design tools and to make the basic system design.	None	None
25.02.2026-27.02.2026	Individually studied Retrieval-Augmented Generation (RAG) architecture and its applicability for domain-specific chatbot systems	Evaluate the feasibility of implementing RAG-based chatbot functionality	Technical complexity of RAG architecture	Reviewed technical documentation and implementation tutorials
28.02.2026	Performed comparative analysis of existing student support platforms and university chatbot systems	Identify features and limitations of existing solutions	Limited availability of detailed system architecture information	Analyzed publicly available documentation and case studies
01.03.2026	Create a basic color palette, design button styles and start designing login screen layout.	To create the first user interface screen.	Layout felt too crowded.	Simplified design by removing unnecessary features.
Summary for the week	This week's main priorities were requirement analysis and individual learning about Figma and RAG architecture. Designed reusable button components and completed login screen design.			
Task for the next week	Begin user interface design and develop low-fidelity wireframes for the Smart UniMate platform.			
Remarks				
Signature of the Student			Date	16.03.2026
Signature of the Mentor			Date	16.03.2026

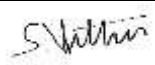
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Week 3

Duration	Start Date	02.03.2026	End Date	08.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
02.03.2026-03.03.2026	Helped team members understand Figma features.	Ensure everyone can use the tool effectively.	Some members struggled with layouts.	Conducted 1 hour Figma training session to the team.
04.03.2026-05.03.2026	Continued development of wireframes for additional system pages.	To design the main pages including dashboard, navigation menu, and information sections.	None	None
06.03.2026	Presented the developed wireframe to the team and mentor.	To let mentor and team to review wireframe content.	None	None
07.03.2026	Conducted team review session for all developed wireframes.	To evaluate the usability and structure of the designed interfaces	Minor disagreements regarding certain UI elements.	Changes were made based on usability and simplicity.
08.03.2026	Finalized the initial wireframe design for the Smart UniMate website.	To complete the early stage UI design before moving to the development phase.	Ensuring all required system features were represented in the wireframes.	Cross-checked wireframes with the requirement list to confirm coverage.
Summary for the week	Partially completed wireframe of our website.			
Task for the next week	I will focus on designing the UI/UX and wireframes for the mobile application to ensure a user-friendly and responsive interface before development.			
Remarks				
Signature of the Student			Date	16.03.2026
Signature of the Mentor			Date	16.03.2026

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Week 4

Duration	Start Date	09.03.2026	End Date	15.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
09.03.2026	Started designing low-fidelity wireframes for the mobile application using Figma.	To design the basic structure of the mobile interface.	None	None
10.03.2026	Designed key mobile screens including home page, chatbot interface, and navigation menu.	Ensure the mobile interface supports main system features.	Some components appeared cluttered on small screens.	Adjusted spacing and simplified navigation elements.
11.03.2026	Developed UI components such as buttons, icons, and card layouts for mobile design.	Maintain design consistency across all screens.	Inconsistent styling between screens.	Created reusable UI components in Figma.
12.03.2026	Conducted internal team review for the mobile UI wireframes.	Gather feedback to improve usability.	None	None
13.03.2026	Presented the mobile UI/UX design to the mentor for feedback.	Validate the design approach before moving to development.	None	None
14.03.2026	None	None	None	None
15.03.2026	Got the feedback from the mentor	None	None	None
Summary for the week	This week, our team finalized and submitted the project journal report to our mentor.			
Task for the next week	Starting the project			
Remarks				
Signature of the Student			Date	16.03.2026
Signature of the Mentor			Date	16.03.2026

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Week 5

Duration	Start Date	16.03.2026	End Date	22.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
16.03.2026	Set up development environment for frontend (VS Code, Node.js, project initialization)	Prepare environment to start frontend development	Faced issues with Node.js version compatibility	Installed correct LTS version and reconfigured environment
17.03.2026	Studied basics of Next.js and React components	Understand frontend framework structure	Difficulty understanding component structure	Watched tutorials and practiced small examples
18.03.2026	Created project structure and folder organization	Maintain clean and scalable code structure	-	-
19.03.2026	Implemented basic layouts	Start building UI structure of the system	Alignment issues in layout	Used tailwind CSS
20.03.2026	-	-	-	-
21.03.2026	-	-	-	-
22.03.2026	Developed login page UI	Start building the project	-	-
Summary for the week	This week focused on transitioning from the design phase to the initial development phase. The frontend development environment was successfully set up, and basic knowledge of Next.js and React was acquired			
Task for the next week	Continue frontend development by implementing additional pages and integrating reusable UI components.			
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

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Week 6

Duration	Start Date	23.03.2026	End Date	29.03.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
23.03.2026	Continued learning advanced React concepts	Improve frontend development skills	None	None
24.03.2026	Developed homepage UI	Build main landing page of the system	None	None
25.03.2026	Implemented chatbot UI interface (frontend only)	Prepare UI for chatbot interaction	UI alignment issues	Adjusted component structure
26.03.2026	Created reusable components (buttons, cards, input fields)	Improve code reusability and consistency	Inconsistency	Standardized styles using CSS
27.03.2026	Reviewed frontend progress with team and members	Enhance overall frontend quality	Feedback required UI Improvements	Updated design based on feedback
28.03.2026				
29.03.2026				
Summary for the week	During this week, the frontend development progressed significantly with the implementation of key pages such as the homepage and chatbot interface. Reusable components were created to ensure consistency across the application. The team also reviewed the progress and made improvements based on feedback.			
Task for the next week	Start integrating frontend with backend services			
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

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Week 7

Duration	Start Date	30.03.2026	End Date	05.04.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
30.03.2026	Reviewed Laravel documentation and set up the Laravel development environment	Begin backend development using Laravel framework	Version conflicts with PHP	Downgraded to compatible PHP version and reinstalled dependencies
31.03.2026	Studied Laravel routing, controllers, and MVC architecture	Understand how Laravel handles requests and responses	None	None
01.04.2026	Created Laravel project structure and configured .env file for database connection	Establish backend database connection	Database connection errors due to incorrect credentials	Corrected .env settings and enabled required PHP extensions
02.04.2026	Reviewed existing Next.js frontend codebase to identify API integration points	Plan frontend-backend connection approach	None	None
03.04.2026	Attended Friday morning mentor meeting to discuss backend integration plan and get guidance	Receive mentor feedback on Laravel and API integration approach	None	None
04.04.2026	None	None	None	None
05.04.2026	None	None	None	None
Summary for the week	This week focused on re-learning Laravel and setting up the backend development environment			
Task for the next week	Create Laravel API endpoints and connect them with the Next.js frontend.			
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

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Week 8

Duration	Start Date	06.04.2026	End Date	12.04.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
06.04.2026	Created RESTful API routes and controllers in Laravel for user authentication	Build backend API endpoints to support login functionality	None	None
07.04.2026	Tested API endpoints using Postman to verify correct responses	Ensure backend APIs return expected data before frontend connection	Some endpoints returned 500 errors	Fixed missing middleware and corrected route definitions
08.04.2026	Configured CORS settings in Laravel to allow requests from the Next.js frontend	Enable cross-origin communication between frontend and backend	CORS policy blocking requests	Added proper CORS headers and configured allowed origins
09.04.2026	None	None	None	None
10.04.2026	Integrated Laravel API with Next.js frontend using Axios for HTTP requests	Establish data communication between frontend and backend	Axios request failures due to CORS misconfiguration	Updated Laravel CORS config and Axios base URL settings
11.04.2026	Attended Friday morning mentor meeting and presented backend-frontend integration progress	Receive feedback on integration approach and plan next steps	None	None
12.04.2026	None	None	None	None
Summary for the week	This week focused on connecting the Laravel backend to the Next.js frontend. RESTful API endpoints were created and tested via Postman.			
Task for the next week	Implement authentication logic in the login page and develop the login API on the backend.			
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 9

Duration	Start Date	13.04.2026	End Date	19.04.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

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Week 10

Duration	Start Date	20.04.2026	End Date	26.04.2026
Date	Task Completed	Objectives of the tasks	Problems	Solutions
20.04.2026	Designed and implemented the login API endpoint in Laravel with input validation rules	Create a secure backend authentication endpoint	None	None
21.04.2026	Implemented Laravel Sanctum for API token-based authentication	Secure the API using token authentication	Sanctum setup required additional configuration steps	Followed official documentation and resolved token mismatch issues
22.04.2026	Connected the Next.js login page to the Laravel authentication API	Enable users to log in through the frontend using backend authentication	Auth token not persisting after login	Stored token in localStorage and configured Axios authorization headers
23.04.2026	None	None	None	None
24.04.2026	None	None	None	None
25.04.2026	Attended Friday morning mentor meeting to review authentication progress and implementation	Receive feedback on login implementation and plan further improvements	None	None
26.04.2026	Met with internal supervisor to discuss overall project progress and remaining milestones	Ensure project is on track and address any scope concerns	None	None
Summary for the week	This week centered on implementing authentication.			
Task for the next week	Apply UI improvements to the login page and implement protected route handling in the frontend.			
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 11

Duration	Start Date		End Date	
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 12

Duration	Start Date		End Date	
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 13

Duration	Start Date		End Date	
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 14

Duration	Start Date		End Date	
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Week 15

Duration	Start Date		End Date	
Date	Task Completed	Objectives of the tasks	Problems	Solutions
Summary for the week				
Task for the next week				
Remarks				
Signature of the Student			Date	
Signature of the Mentor			Date	

Mentor/(s) Approval for the Final Submission

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Internal Supervisor/(s) Approval for the Final Submission

[illegible]

Any other Remarks

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